

Day 2 — Transformers Interview Revision

Q/K/V + Scaled Dot-Product Attention

High-Yield Concepts

- Query, Key, and Value
- Learned linear projections
- $QK^T \rightarrow$ attention scores
- Scaling by $\sqrt{d_k}$
- Softmax $\rightarrow$ attention weights
- Weighted sum of Value vectors
- Complete scaled dot-product attention pipeline

Core Formulas

$$Q = XW_Q$$

$$K = XW_K$$

$$V = XW_V$$

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$$

Interview Questions + Answers

1. What are Query, Key, and Value?

Q, K, and V are learned linear projections of the input token representations. Q and K determine compatibility or relevance, while V contains the information that is actually aggregated.

2. How are Q, K, and V generated?

Each input representation is multiplied by its own trainable weight matrix: $Q = XW_Q$, $K = XW_K$, and $V = XW_V$. These are learned linear projections.

3. What does QK^T calculate?

It calculates the dot product between every Query and every Key. The (i,j) entry measures how relevant token j is to token i based on their learned Query and Key representations.

4. Why divide by $\sqrt{d_k}$?

As dimensionality increases, dot products can become large in magnitude. Large scores make softmax extremely peaked and can produce very small gradients. Scaling keeps the scores at a reasonable range and makes training more stable.

5. Why use softmax?

Softmax converts the scaled attention scores into positive, normalized attention weights that sum to one. These weights indicate how much each Value should contribute to the output.

6. Why multiply the attention weights by V?

The attention weights determine how much information to take from each Value vector. The weighted sum creates the new representation for each token.

Dimension Walkthrough

Using the standard row-token convention: X is $N \times D$, where N is the number of tokens and D is the embedding dimension. W_Q , W_K , and W_V are $D \times P$, where P is the Q/K/V dimension.

Operation	Dimensions	Meaning
$Q = XW_Q$	$(N \times D)(D \times P) = N \times P$	Query representations
$K = XW_K$	$(N \times D)(D \times P) = N \times P$	Key representations
$V = XW_V$	$(N \times D)(D \times P) = N \times P$	Value representations
$QK^\top$	$(N \times P)(P \times N) = N \times N$	Score for every token pair
$A = \text{softmax}(QK^\top / \sqrt{d_K})$	$N \times N$	Normalized attention weights
AV	$(N \times N)(N \times P) = N \times P$	Final attention output

Column-Vector Convention

If token representations are stored as columns instead of rows, V can be represented as $P \times N$. Then the weighted combination can be written as $V A$, producing $P \times N$. Transposing gives the conventional $N \times P$ orientation. This is simply a matrix-orientation convention; the underlying attention computation is the same.

Interview-Ready Summary

Q and K determine where to attend; V provides the information to aggregate. $QK^\top$ produces pairwise relevance scores. Scaling prevents large dot products from making softmax too peaked. Softmax converts the scores into normalized attention weights, and those weights are used to calculate a weighted sum of the Value vectors.